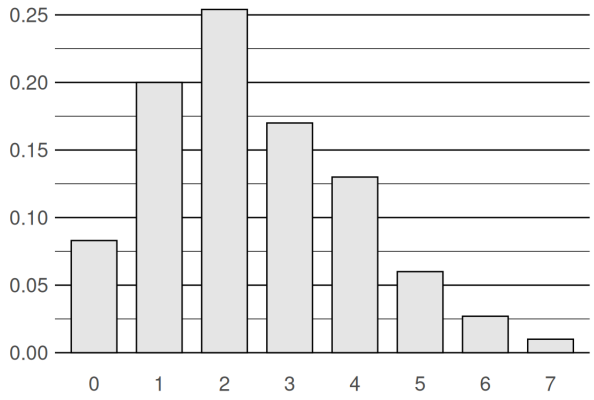
4 Probability

1. The number of goals in a game in the FIFA World Cup is given by the random variable X. The distribution of X is given in the plot below.



(a) (3 points) Find the 25th, 50th and the 75th percentile of X.

(b) (3 points) Is $E(X)$ equal to, smaller than, or larger than the 50th percentile of X? Explain how you can tell without knowing $E(X)$.

(c) (3 points) Which option is closest to the probability of scoring 5 or more goals in a FIFA World Cup game? **Select ONE**.

◯ 0.11 ◯ 0.02
◯ 0.5 ◯ 0.96

2. The distribution of the number of children in a household (X) in a certain neighborhood is given below.

x	0	1	2	3
$P(X = x)$	?	0.4	0.25	0.15

It can be shown that $E(X) = 1.35$ and $SD(X) = 0.96$.

(a) (3 points) Which of the following statements is the most specific accurate description of X? **Select ONE**.

- ◯ X is a discrete RV.
- ◯ X is a binomial RV.
- ◯ X is a continuous RV.
- ◯ X is a normal RV.

(b) (3 points) What is the median number of children per household? **Select ONE**.

- ◯ 0
- ◯ 1
- ◯ 2
- ◯ Cannot be determined from the information given

(c) (3 points) Consider taking a random sample of four households from this neighborhood. What is the distribution of $\bar{X}_4$, the average number of children per household?

- ◯ $\bar{X} \dot{\sim} N(1.35, 0.96)$
- ◯ $\bar{X} \dot{\sim} N(1.35, 0.96\sqrt{4})$
- ◯ $\bar{X} \dot{\sim} N(1.35, \frac{0.96}{\sqrt{4}})$
- ◯ None of the above

3. Researchers have planted wildflowers in several plots in an attempt to study local insect wildlife. Each day, the number of monarch butterflies in each plot is observed. Suppose the number of monarch butterflies per plot is given by random variable X with the following partial probability distribution:

x	0	1	2	3
$P(X = x)$	?	?	0.2	0.1

The average number of monarch butterflies per plot is $E(X) = 0.8$.

(a) (4 points) Find $P(X = 0)$ and $P(X = 1)$.

(b) (3 points) Write an expression for $\text{Var}(X)$. You do not need to simplify your answer and can leave it as an expression.

4. The number of credits that a UW-Madison course is worth follows this distribution, with some numbers removed. (Not real data)

x	1	2	3	4
$P(X = x)$	?	?	0.4	0.4

(a) (3 points) Suppose $P(X = 1)$ and $P(X = 2)$ are equal. What are $P(X = 1)$ and $P(X = 2)$?

(b) (3 points) Write out a mathematical expression for $E[X]$. You do not need to simplify your answer and can leave it as an expression.

(c) (3 points) Write out a mathematical expression for $Var[X]$. You can write $E[X]$ instead of the full expression above where needed. You do not need to simplify your answer and can leave it as an expression.

5. Based on historic data, a carnival believes that a player has a 7% chance of winning at a carnival basketball game. They model the number of winners (out of 350 total players) as Binom(350, 0.07).

   (a) (5 points) Identify the four assumptions that are being made for this binomial model, and briefly discuss how each one applies to this situation. Name one assumption that might not be well met, and explain.

   (b) Assuming the binomial model is accurate, fill in the three blanks of `pbinom()` below to calculate the probability that 30 or more players out of 350 win basketball.

```
1 - pbinom(_____, _____, _____)
```

   (c) Consider performing the calculation in (b) with a normal approximation. Fill in the four blanks of `pnorm()` below to approximate the probability that 30 or more players out of 350 win basketball.

   You don't need to simplify your answers for the blanks.

```
pnorm(_____, _____, _____,
      _____)
```

6. Four different bird species, with different levels of rarity, live in a conservation area. Scientists regularly walk through the area and record the count of unique of bird species they observe (out of 4). A distribution for X, the number of species observed, is given below.

x	0	1	2	3	4
$P(X = x)$	0.2	0.4	0.25	0.1	0.05

(a) (5 points) For each of the four BINS assumptions, comment on whether the assumption is met or not met for X. Is X a binomial random variable?

(b) (3 points) What is the 75th percentile of X?

(c) (3 points) Write an expression to calculate $E(X)$. You do not need to simplify your answer and can leave it as an expression.

7. Define four different independent binomial random variables.

$$X_1 \sim \text{Binom}(5, 0.6), \qquad X_2 \sim \text{Binom}(25, 0.4), \qquad X_3 \sim \text{Binom}(100, 0.1), \qquad X_4 \sim \text{Binom}(100, 0.4)$$

(a) (4 points) Fill in the blanks below with “less than", “greater than", or “equal to".

- The expectation of X_1 is _____ 2.5.
- The smallest possible value of X_1 is _____ the smallest possible value of of X_4.
- The expectation of X_3 is _____ the expectation of X_4.
- The standard deviation of X_3 is _____ the standard deviation of X_4.

(b) (3 points) Which of the resulting combinations below is also a binomial random variable? **Select ONE**.

◯ $Y_1 = X_1 + X_2$ ◯ $Y_3 = X_2 + X_4$
◯ $Y_2 = X_2 + X_3$ ◯ $Y_4 = X_3 + X_4$

(c) (3 points) Which of the four variables is *least* well approximated as $N\left(np, \sqrt{np(1-p)}\right)$? **Select ONE**.

◯ X_1 ◯ X_3
◯ X_2 ◯ X_4

8. 32 men competed in the 2024 Paris Qualification Round for shot put. If an athlete threw the ball at least 21.35 meters, they qualified for the final.

   (a) (5 points) Let X be the number of athletes who qualify for the final. Determine whether X meets the assumptions of a binomial random variable by briefly explaining why each individual assumption is or is not met.

   (b) (4 points) Regardless of your answer to the previous question, assume that X, the number of athletes who qualify for the final, follows a binomial distribution, with some number of trials n and some probability of success p.

   Which lines of R code below can be used to calculate the probability that 10 or more athletes qualify for the final? **Select ALL that apply**.

   ◯ `pbinom(9, n, p)`
   ◯ `1 - pbinom(9, n, p)`
   ◯ `1 - pbinom(10, n, p)`
   ◯ `pbinom(10, n, p)`

9. (a) (3 points) For the three random variables listed below, identify its distribution, which will be either binomial or normal. Be sure to write which *specific* binomial or normal RV it follows.

   i. A multiple-choice test has 15 questions, each of which has five choices. A specific unprepared student takes the test and has a 80% chance of getting a question wrong. Let X_1 denote the number of answers that the student gets right in the test. Find the distribution of X_1.

   ii. Suppose that grade points of undergraduate students at a university have a distribution with a mean of 2.63 and a variance of 0.15. Let X_2 denote the average grade points of 100 undergraduate students. Find the distribution of X_2.

   iii. An experimental medication was given to 3000 patients with a certain medical condition. It is previously known that 2.6% people show severe side effects to a component in the medication. Let X_3 denote the number of patients who develop severe side effects. Find the distribution of X_3.

   (b) (3 points) Out of the variables binomially distributed in (a), which of those can be most accurately approximated by a normal distribution? What will be the normal approximation for that variable?

10. Let $M \sim N(10, 1)$ denote the shoe sizes for males in the U.S. The area under the curve between shoe size 8 and 12 is 95%, i.e. $P(8 \leq M \leq 12) = 0.95$.

    Let $F \sim N(8, 0.8)$ denote the shoe sizes for females in the U.S.

    (a) (3 points) Which is taller - the height of the peak of the M bell-curve or the height of the peak of the F bell-curve? Briefly explain your answer.

    (b) (4 points) Find the points lo and hi such that the area under the curve for F is 95%. In other words, $P(lo \leq F \leq hi) = 0.95$. You do not need to simplify your answers and can leave them as an expression.

    (c) (3 points) We want to confirm that the values `lo` and `hi` from part (a) actually do have 95% of the area between them. Which lines of R code return the area between `lo` and `hi`? **Select ALL that apply**.

    ◯ `2*pnorm(lo, 8, 0.8)`
    ◯ `2*pnorm(hi, 8, 0.8)`
    ◯ `1 - 2*pnorm(lo, 8, 0.8)`
    ◯ `1 - 2*pnorm(hi, 8, 0.8)`
    ◯ `pnorm(lo, 8, 0.8) - pnorm(hi, 8, 0.8)`
    ◯ `pnorm(hi, 8, 0.8) - pnorm(lo, 8, 0.8)`

11. Consider a standard normal variable $Z \sim N(0,1)$. The table below lists several lower-tail probabilities of Z.

Z-score	$P(Z \leq -1.5)$	$P(Z \leq -1)$	$P(Z \leq -0.5)$	$P(Z \leq 0.5)$	$P(Z \leq 1)$	$P(Z \leq 1.5)$
`pnorm`	`pnorm(-1.5)`	`pnorm(-1)`	`pnorm(-0.5)`	`pnorm(0.5)`	`pnorm(1)`	`pnorm(1.5)`
Probability	0.067	0.159	0.309	0.691	0.841	0.933

Use the table of probabilities to answer the questions below.

(a) (3 points) What is $P(-1 \leq Z \leq 0.5)$? **Select ONE**.

- ◯ 0.691
- ◯ $0.691 - 0.159$
- ◯ $0.691 + 0.159$
- ◯ $1 - (0.691 + 0.159)$

(b) (3 points) What is $P(Z \leq -1 \text{ OR } Z > 1)$? **Select ONE**.

- ◯ 0
- ◯ 0.159
- ◯ 2×0.159
- ◯ 0.691
- ◯ 2×0.691
- ◯ 1

(c) (3 points) What value is closest to the 95th percentile of Z? **Select ONE**.

- ◯ -1.5
- ◯ -1
- ◯ -0.5
- ◯ 0.5
- ◯ 1
- ◯ 1.5

12. Assume that the monthly electricity consumptions of all households in a certain region is approximately normally distributed with a mean of 1200 kilowatt-hours and a standard deviation of 110 kilowatt-hours.

    (a) (3 points) John Smith recieved a notice informing him that his household electricity consumption lies 2 standard deviation above the mean. What is John Smith's household electricity consumption? You do not need to simplify your answer and can leave it as an expression.

    (b) (3 points) Find the value of average electricity consumption in 100 houses such that it lies 2 standard deviation below the mean of the *sampling distribution of sample means*. You do not need to simplify your answer and can leave it as an expression.

    (c) (4 points) Using the *d*, *p* or *q* function for the normal distribution, fill in the blanks below such that you get R code for the interquartile range of the monthly electricity consumptions of all households in that region.

    The interquartile range is the difference between the 25th percentile and the 75th percentile of the data/distribution, or the range of the middle 50% of the data/distribution.

    IQR = _____(_____, 1200, 110) -
    _____(_____, 1200, 110)

13. You are interested in comparing the performance of Student A at University A with Student B at University B.

Grade point averages at University A are normally distributed with mean 2.8 and standard deviation 0.4. Grade point averages at University B are normally distributed with mean 2.5 and standard deviation 0.5.

(a) (3 points) Student A has a GPA of 3.2 at University A. Which R code below calculates the GPA of Student B who has the same percentile GPA at University B? **Select ONE**.

- ◯ `pnorm(3.2, 2.8, 0.4) %>% qnorm(2.5, 0.5)`
- ◯ `pnorm(3.2, 2.5, 0.5) %>% qnorm(2.8, 0.4)`
- ◯ `qnorm(3.2, 2.8, 0.4) %>% pnorm(2.5, 0.5)`
- ◯ `qnorm(3.2, 2.5, 0.5) %>% pnorm(2.8, 0.4)`

(b) (3 points) What GPA score from University A corresponds to a z-score of -1? **Select ONE**.

- ◯ -0.4
- ◯ 0.4
- ◯ 2.4
- ◯ 2.8

(c) (3 points) Consider taking a random sample of five students from University A. What is the distribution of $\bar{X}$, the average of their GPA's? **Select ONE**.

- ◯ $\bar{X} \dot{\sim} N(2.8, 0.4)$
- ◯ $\bar{X} \dot{\sim} N(2.8, 0.4\sqrt{5})$
- ◯ $\bar{X} \dot{\sim} N(2.8, \frac{0.4}{\sqrt{5}})$
- ◯ None of the above

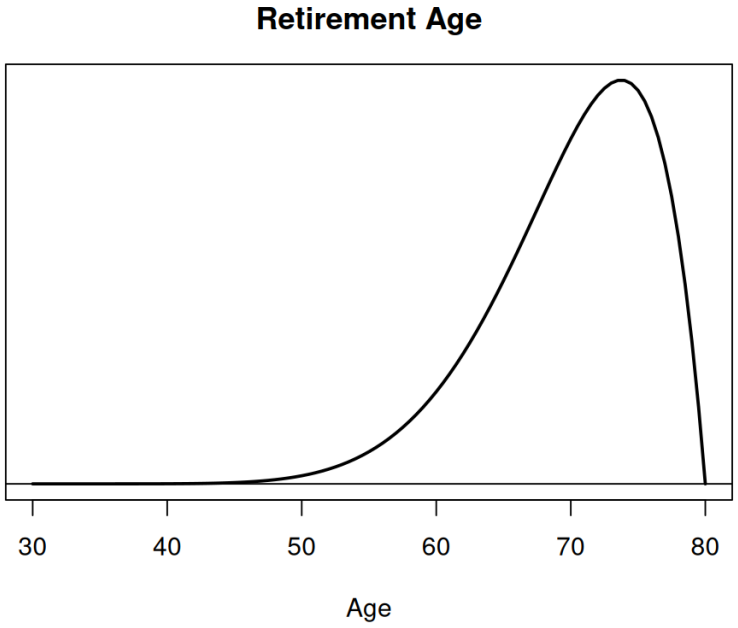
14. At a certain hospital, the number of liters of blood donated per week is given by normal variable D with $\mu_D = 51, \sigma_D = 7$. The number of liters of blood required for transfusions per week is given by normal variable R with $\mu_R = 56, \sigma_R = 4$. You can assume R and D are independent.

(a) (4 points) The number of liters required this week is $r = 48$. What number of donated liters has the same percentile as $r = 48$? In other words, if $P(R \leq 48) = p$, find d such that $P(D \leq d) = p$.

(b) (3 points) Identify the sampling distribution for $\bar{D}_{52}$, the average liters of blood donated per week over a 52-week year.

$$\bar{D}_{52} \sim _____(_____, _____)$$

15. The continuous random variable X, shown below, represents the population of retirement ages in the US. X is defined on [30, 80] and has has expected value 64 and variance 92.16.



(a) (3 points) **Select ONE**. X will realize to a value greater than 64:

- ◯ less than half the time
- ◯ close to half the time
- ◯ more than half the time

(b) (3 points) **Select ONE**. The sample mean $\bar{X}_{75}$ is calculated by taking the average of 75 random draws from X. $\bar{X}_{75}$ will realize to a value greater than 64:

- ◯ less than half the time
- ◯ close to half the time
- ◯ more than half the time

(c) (5 points) Fill in the blanks to write R code to approximate $P(\bar{X}_{75} \geq 62)$. Your answer should be a function with four arguments.

_____(_____, _____, _____,
_____)

16. The plot below shows the densities of three different distributions, named Solid, Dashed, and Dotted. One of these distributions is $N(0,1)$, standard normal, another one is $t(5)$, the t distribution with 5 degrees of freedom, and the final one is $D(0,1)$, some other distribution with mean 0 and standard deviation 1.



(a) (3 points) Identify each distribution by the corresponding line type.

$N(0,1)$:	◯ Solid	◯ Dashed	◯ Dotted
$t(5)$:	◯ Solid	◯ Dashed	◯ Dotted
$D(0,1)$:	◯ Solid	◯ Dashed	◯ Dotted

(b) (3 points) Rank the three distributions using their line type names (Solid, Dashed, Dotted) in order from that with the smallest to the largest values of the 0.975 quantile.

(Smallest) _____ < _____ < _____ (Largest)

(c) (3 points) A random sample of size $n = 400$ is drawn from distribution $D(0,1)$ and we calculate the sample mean

$$\bar{X} = \frac{1}{400}\sum_{i=1}^{400} X_i.$$

What are the mean and standard deviation of the distribution of $\bar{X}$? **Select ONE**.

◯ $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = 1$

◯ $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = \frac{1}{400}$

◯ $\mu_{\bar{X}} = 400, \quad \sigma_{\bar{X}} = 400$

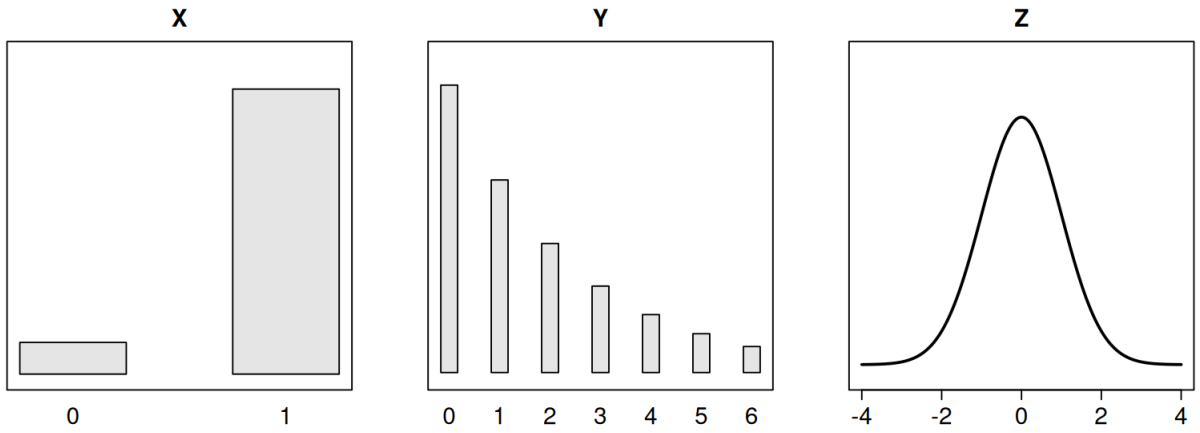
◯ $\mu_{\bar{X}} = 0, \quad \sigma_{\bar{X}} = \frac{1}{\sqrt{400}}$

(d) (3 points) Which of the three distributions will have a shape closest to that of the distribution of $\bar{X}$? (Note that the scale will be different.) **Select ONE**.

◯ $N(0,1)$ ◯ $t(4)$ ◯ $D(0,1)$

5 Inference

1. Consider the following random variables: X is discrete and can take on values 0 or 1, Y is a different skewed discrete variable, and Z is normal.



(a) (5 points) Which of the following will have a normal or approximately normal distribution? **Select ALL that apply**.

- ◯ The average of 200 samples from X.
- ◯ The average of 100 samples from Y.
- ◯ The median of 100 samples from Y.
- ◯ The average of 1 sample from Z.
- ◯ The sum of 200 samples from X.

A researcher who does not know the true distribution of X wants to estimate $p = P(X = 1)$. They report a 95% confidence interval for p as $(0.845, 0.921)$.

(b) (3 points) Which line of R code correctly calculates the critical value / quantile score for the 95% CI? **Select ONE**.

- ◯ `qnorm(0.9)`
- ◯ `qnorm(0.95)`
- ◯ `qnorm(0.975)`
- ◯ `qnorm(0.99)`

(c) (4 points) Suppose the researcher instead set $\alpha = 0.02$ to determine the confidence level. Assuming the data does not change, how does this interval compare to the 95% CI above? Fill in the blanks with “a smaller", “the same", or “a larger".

The new CI would have _____ margin of error compared to the 95% CI.

The new CI would have _____ point estimate compared to the 95% CI.

2. A doctor records the cholesterol level of 20 of her patients (in mg/dL). She reports a 95% confidence interval for the average cholesterol:

$$(147.29, 215.71)$$

(a) (3 points) Identify the point estimate and margin of error of this CI. You do not need to simplify your answer.

(b) (3 points) Which of the following statements is the best interpretation of this CI? **Select ONE**.

- ◯ If we repeated the CI procedure across multiple samples, 95% of the intervals would cover the cholesterol level for the patients in the sample.
- ◯ If we repeated the CI procedure across multiple samples, 95% of the intervals would cover the mean cholesterol level for all patients.
- ◯ 95% of the 20 patients in this sample have a cholesterol level between 147.29 and 215.71.
- ◯ 95% of all patients have a cholesterol level between 147.29 and 215.71.

(c) (3 points) The doctor wants to be especially sure that her confidence interval correctly covers the true mean cholesterol of her patients. What change should she make to the 95% interval above? **Select ONE**.

- ◯ Choose a smaller confidence level, which gives a narrower interval.
- ◯ Choose a larger confidence level, which gives a narrower interval.
- ◯ Choose a smaller confidence level, which gives a wider interval.
- ◯ Choose a larger confidence level, which gives a wider interval.

3. A 95% confidence interval for the true proportion of left-handed Americans is reported as (0.086, 0.112).

   (a) (3 points) Which line of R code correctly calculates the critical value / quantile score for the margin of a 95% CI? **Select ONE**.

   ◯ `qnorm(0.05)` ◯ `qnorm(0.95)`
   ◯ `qnorm(0.9)` ◯ `qnorm(0.975)`

   (b) (4 points) Which of the following statements are true about the CI? **Select ALL that apply**.

   ◯ A 98% CI on the same data would be narrower than the 95% CI.
   ◯ A 98% CI on the same data would be wider than the 95% CI.
   ◯ A 98% CI would have the same point estimate as the 95% CI.
   ◯ A 98% CI would have a different point estimate as the 95% CI.

   (c) (3 points) A hypothesis test of $H_0 : p = 0.1$ versus $H_A : p \neq 0.1$ (where p is the true proportion of left-handed Americans) is performed. The test resulted in a p-value smaller than the chosen significance level $\alpha = 0.05$. Which of the following statements is the best conclusion of this test? **Select ONE**.

   ◯ The true proportion of left-handed Americans is 0.1.
   ◯ The true proportion of left-handed Americans is not 0.1.
   ◯ We do not have evidence that the true proportion of left-handed Americans is different from 0.1.
   ◯ We have evidence that the true proportion of left-handed Americans is different from 0.1.

4. We are interested in the true percentage, p, of UW-Madison students who know that Bucky Badger's full first name is Buckingham. We ask 100 students, and construct a 90% confidence interval for p as (0.50, 0.70).

   (a) (3 points) Which of the following correctly calculates the critical value for a 90% CI? **Select ONE**.

   ◯ `qnorm(0.9)`
   ◯ `qnorm(0.95)`
   ◯ `qnorm(0.975)`
   ◯ `dbinom(2, 3, 0.4)`

   (b) (4 points) Which of the following statements are TRUE about this interval? **Select ALL that apply**.

   ◯ A 95% confidence interval would have a larger margin of error than this one.
   ◯ A 95% confidence interval would have the same margin of error as this one.
   ◯ If we had asked 500 students AND decreased confidence level to 80%, the resulting interval would be narrower than this one.
   ◯ If we had asked 500 students AND decreased confidence level to 80%, we do not know if the resulting interval would be narrower or wider than this one.

   (c) (3 points) Which of the following conclusions can be made from this confidence interval? **Select ONE**.

   ◯ At least half of all UW-Madison students know this fact.
   ◯ p is below 0.8.
   ◯ 60% of the students sampled knew this fact.
   ◯ We are 95% confident that 60% of all UW students know this fact.

5. Carabiners made at a factory must have a breaking strength of at least 4500 lbs ($H_0 : \mu \geq 4500$). A manager thinks that the mean breaking strength may be less than 4500 lbs ($H_A : \mu < 4500$) so he takes a sample of carabiners to test his hypothesis with $\alpha = 0.1$. If he has evidence that $\mu < 4500$, he will halt carabiner production.

   (a) (5 points) Classify each scenario as a type I error, type II error, or correct decision. **Select ONE each.**

      i. The true mean breaking strength is 4500 lbs. The manager allows production to continue.

      ◯ Type I error ◯ Type II error ◯ Correct decision

      ii. The true mean breaking strength is 4500 lbs. The manager halts production.

      ◯ Type I error ◯ Type II error ◯ Correct decision

      iii. The true mean breaking strength is 4300 lbs. The manager allows production to continue.

      ◯ Type I error ◯ Type II error ◯ Correct decision

      iv. The true mean breaking strength is 4300 lbs. The manager halts production.

      ◯ Type I error ◯ Type II error ◯ Correct decision

      v. The true mean breaking strength is 4800 lbs. The manager allows production to continue.

      ◯ Type I error ◯ Type II error ◯ Correct decision

   (b) (3 points) Explain whether the manager should make α larger or smaller to avoid mistakenly selling carabiners that are not strong enough.

6. A manager at a campus cafeteria is deciding what pizza toppings to offer. He thinks that if over 25% of his customers like pineapple on pizza, he will offer pineapple as a topping.

   He takes a sample of 45 customers and finds that 14 of them like pineapple on pizza.

   (a) (3 points) Identify null and alternative hypotheses to answer the manager's question. They should be in terms of p, the proportion of customers who like pineapple on pizza.

   (b) (3 points) The manager will model the number of customers who like pineapple as Binom(45, p). Which line of R code below correctly calculates the p-value for the test? **Select ONE**.

   - ◯ `pbinom(14, 45, 0.25)`
   - ◯ `1 - pbinom(14, 45, 0.25)`
   - ◯ `pbinom(14, 45, 14/45)`
   - ◯ `1 - pbinom(13, 45, 0.25)`
   - ◯ `1 - pbinom(15, 45, 0.25)`
   - ◯ `1 - pbinom(15, 45, 14/45)`

   (c) (5 points) The observed p-value is greater than the manager's chosen significance level $\alpha = 0.05$. Which of the following statements are true? **Select ALL that apply**.

   - ◯ We are certain that 25% or less of all customers like pineapple on pizza.
   - ◯ It is possible that more than 25% of all customers like pineapple on pizza.
   - ◯ We do not have evidence that more than 25% of all customers like pineapple on pizza.
   - ◯ We could have made a Type I error.
   - ◯ We could have made a Type II error.